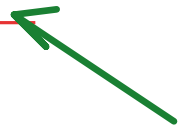
Handout: the RC circuit

A resistor R in series with a capacitor C charges with the time constant tau = R C. After one tau the voltage reaches 63 percent of its final value, after five it has settled.

The rise time from 10 to 90 percent is about 2.2 tau.

~~Doubling R doubles every one of these times.~~



Questions

1. What is tau for R = 2 k and C = 10 n?
2. How long until it settles?

Remember: 2.2 tau

Answers

1. tau = 2000 x 10e-9 = 20 microseconds.
2. About five tau, so 100 microseconds.

Checked on 12 November

Check the result on the oscilloscope with a square wave

whose period is much longer than tau.